

TEMPO: Trajectory EM Mixture of POD Operators

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Problem Statement

Task

Given $\mathcal{D} = \{(u_0^{(i)}, \kappa^{(i)}, s^{(i)})\}_{i=1}^N$, construct $\hat{\mathcal{G}} \approx \mathcal{G}: (u_0, \kappa) \mapsto s$ accurate across all qualitative regimes, without solving the PDE.

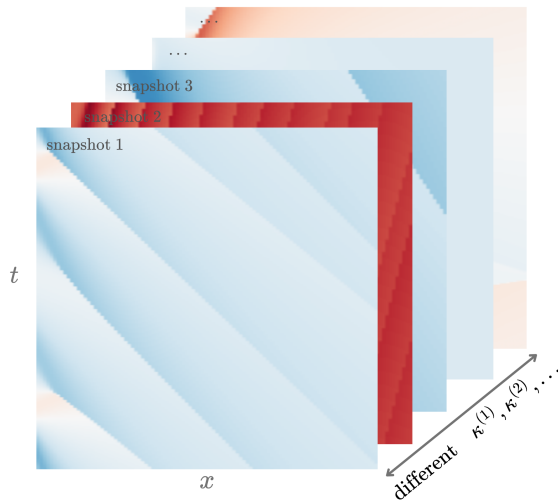
Problem

When solutions change character fundamentally across κ (shocks at $\nu=0.001$ vs. smooth diffusion at $\nu=1.0$), a single global basis cannot fit all regimes well, and error concentrates where it matters most.

Proposal: discover latent regimes and build a dedicated basis per regime.

TEMPO — *Trajectory EM Mixture of POD Operators*

TEMPO: Central Idea



Challenge

A single model fails when solutions change character fundamentally across κ .

Phase 1 — Offline: discover M solution regimes from data; fit a dedicated basis to each

Phase 2 — Online: a gating network routes each new input to the right regime; local branch networks predict the solution:

$$\hat{s} = \sum_m g_m(u_0, \kappa) \left[\phi_0^{(m)} + \sum_k b_{mk} \phi_k^{(m)} \right]$$

Related Work

DeepONet

Lu et al., 2019

Branch encodes input function; trunk encodes query location. Universal approximation of nonlinear operators.

POD-DeepONet

Lu et al., 2022

Fixed POD basis replaces the trunk network. Compact, interpretable, data-efficient.

Neural-POD

Mou et al., 2026

Nonlinear orthogonal basis via greedy residual minimization. Captures structure beyond any linear subspace.

Local Reduced-Order Bases

Amsallem et al., 2012

Snapshot space partitioned into clusters; separate POD basis per cluster. Hard assignment, offline only.

Local Bases + Selector Network

Daniel et al., 2020

Subspace clustering on the Grassmann manifold; dictionary selector at inference. Hard assignment, offline only, disconnected from basis quality.

TEMPO integrates regime discovery and operator learning end-to-end.

Generative Model

Hypothesis. Each trajectory arises from one of M latent regimes; labels z_i are unobserved.

$$z_i \sim \text{Categorical}(\pi_1, \dots, \pi_M)$$

$$(s^{(i)} \mid z_i=m) \sim \mathcal{N}(\hat{s}_m^{(i)}, \sigma^2 W^{-1})$$

$$\hat{s}_m^{(i)}(y) = \phi_0^{(m)}(y) + \sum_{k=1}^{K_m} \lambda_k^{(m,i)} \phi_k^{(m)}(y)$$

z_i regime label (latent)

π_m prior weight

σ^2 assignment softness

$\phi_k^{(m)}$ spatiotemporal mode

$\lambda_k^{(m,i)}$ scalar amplitude

Find $\Phi^*, \pi^*, \sigma^{2*}$ maximising the observed-data log-likelihood (EM).

$$\varepsilon = \frac{1}{N_{\text{test}}} \sum_i \frac{\|s^{(i)} - \hat{s}^{(i)}\|_w}{\|s^{(i)}\|_w}$$

Phase 1 — Basis Learning

Two basis types

POD (linear):

$\tilde{S} \approx U_K \Sigma_K V_K^\top$; modes $\{\phi_k\}$ are columns of U_K

NeuralPOD (nonlinear):

each mode ϕ_k is a Fourier feature MLP;

greedy residual minimization:

$$\min_{\phi_k, \{\lambda_k^{(i)}\}} \sum_{i=1}^N \|r^{(k-1,i)} - \lambda_k^{(i)} \phi_k\|_w^2$$

$r^{(k-1,i)}$ — residual after fitting $k-1$ modes
stops when residual $< \tau$ of initial variance

TEMPO: EM over M regimes

E-step — soft regime assignments:

$$\gamma_{im} \propto \pi_m \exp\left(-\frac{\|s^{(i)} - \hat{s}_m^{(i)}\|_w^2}{2\sigma^2}\right)$$

M-step — γ_{im} -weighted basis per regime:

$$\min_{\Phi_m} \sum_{i=1}^N \gamma_{im} \|s^{(i)} - \hat{s}_m^{(i)}\|_w^2$$

solved by γ_{im} -weighted SVD (POD) or greedy MLP (NeuralPOD)

Better bases $\rightarrow$ sharper assignments $\rightarrow$ better bases.

Phase 2 — Online Operator

Single-regime baselines

Bases $\{\phi_k\}$ frozen from Phase 1.

BranchNet predicts expansion coefficients:

$$\min_{\theta} \frac{1}{N} \sum_{i=1}^N \left\| \mathcal{B}_{\theta}(u_0^{(i)}, \kappa^{(i)}) - \boldsymbol{\lambda}^{(i)} \right\|^2$$

Prediction at any (x, t) :

$$\hat{s} = \phi_0 + \sum_{k=1}^K \mathcal{B}_{\theta,k}(u_0, \kappa) \phi_k$$

Bases $\{\Phi_m^\}$ and responsibilities $\{\gamma_{im}^*\}$ frozen; only GatingNet and BranchNets are trained.*

TEMPO: gated operator

GatingNet $g_m + M$ BranchNets b_{mk} :

$$\hat{s} = \sum_m g_m(u_0, \kappa) \left[\phi_0^{(m)} + \sum_k b_{mk} \phi_k^{(m)} \right]$$

Training loss:

$$\mathcal{L} = \mathcal{L}_{\text{data}} + \alpha_{\text{KL}} \mathcal{L}_{\text{KL}} - \alpha_H \mathcal{L}_{\text{ent}}$$

$\mathcal{L}_{\text{KL}}$: anchors gating to EM responsibilities

$\mathcal{L}_{\text{ent}}$: prevents collapse to a single regime

Computational Experiment

Datasets (PDEBench):

1D Burgers' $u_t + uu_x = \nu u_{xx}, \quad x \in (-1, 1),$
 $t \in (0, 2]$

$\nu \in \{0.001, 0.01, 0.1, 1.0\}$; 9,500 traj. per
 ν ; $N_x=1,024$, $N_t=201$

smooth diffusion at $\nu=1.0 \rightarrow$ near-discontinuous shocks
at $\nu=0.001$

2D Darcy flow $-\nabla \cdot (a \nabla u) = \beta, \quad x \in (0, 1)^2$

$\beta \in \{0.01, 0.1, 1, 10, 100\}$; 8,000 train per
 β ; 128×128 grid

solution scales linearly with β — cross-parameter errors
exceed 10^3

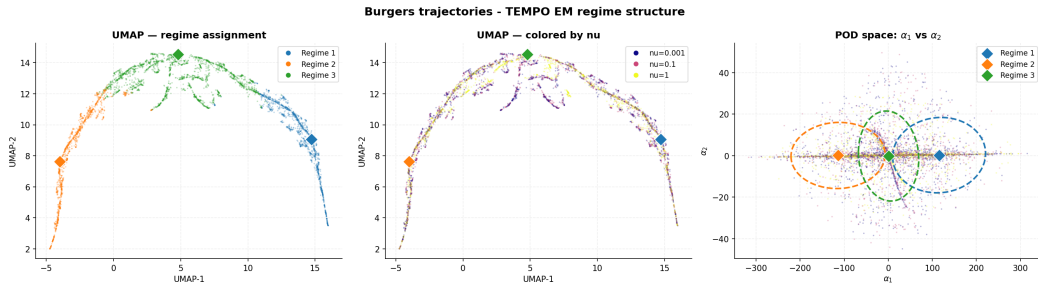
Verification claims:

- 1) EM recovers interpretable regime structure from unlabelled data
- 2) Local bases outperform a single global basis at equal mode count
- 3) TEMPO outperforms a jointly-trained model without regime discovery

Methods:

DeepONet, FNO, CNO	global, no basis
POD-DeepONet	global, linear
NeuralPOD-DeepONet	global, NeuralPOD
TEMPO(POD)	M regimes, linear
TEMPO(NeuralPOD)	M regimes, NeuralPOD

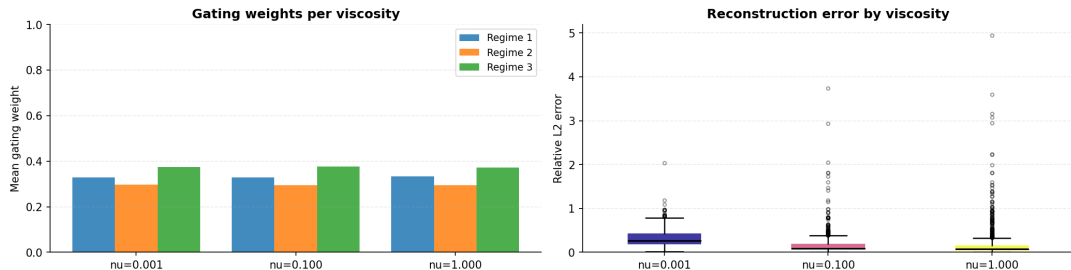
Error Analysis



TEMPO(NeuralIPOD), $M=3$: UMAP coloured by regime (*left*), by ν (*centre*), GMM fit in coefficient space (*right*). Sharper assignments ($H=0.23$) than POD ($H=0.84$).

Results: gating and error

TEMPO Phase 2 - test set results

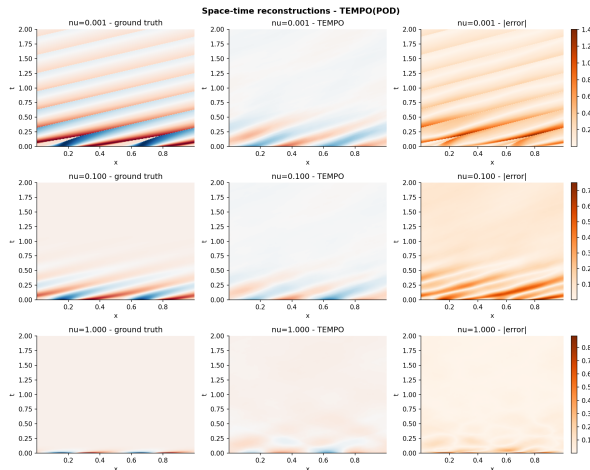


TEMPO(POD), $M=3$, joint training on $\nu \in \{0.001, 0.1, 1.0\}$.

Left: mean gating weight g_m per regime at each ν — all three regimes activate equally across the full viscosity range, confirming that EM partitions by solution geometry, not by ν .

Right: per-trajectory reconstruction error

Results: reconstruction



TEMPO(POD) ($M=3$): ground truth / prediction / error

Joint training on Burgers'
($\nu \in \{0.001, 0.1, 1.0\}$):

	0.001	0.1	1.0
POD-DeepONet	0.460	0.339	0.210
TEMPO(POD)	0.317	0.168	0.182

TEMPO(POD) improves by **13–50%** across all ν simultaneously — a single jointly-trained model without regime discovery cannot match this.

Conclusion

TEMPO: two-phase framework

Offline: EM discovers M latent regimes; γ_{im} -weighted basis (POD or NeuralPOD) per regime

Online: GatingNet anchored to γ_{im}^* via $\mathcal{L}_{\text{KL}}$; M BranchNets predict regime-specific coefficients

Main finding

TEMPO(POD) outperforms per- κ specialists across the full parameter range and the joint POD-DeepONet baseline by 13–50% on Burgers', up to $3\times$ on Darcy flow. Discovered regimes cross κ isolines — geometry-driven rather than parameter-driven.

Open directions

Parameter-conditioned modes $\phi_k^{(m)}(\cdot; \kappa)$

Extension to higher-dimensional benchmarks (Navier–Stokes)

Convergence theory for Generalized EM with nonlinear bases